

Securing Data in IoT Constrained Node Networks

This is the second part of the blog series, 'IoT Constrained Node Networks'. Reading the first part, 'An Introduction to IoT Constrained Node Networks', which was carried in the November 2019 issue of *OSFY*, will ensure a better understanding of this article.



This blog provides an overview of cyber security standards, tells how to use the NIST cyber security framework, and presents some guidelines to secure data in IoT constrained node networks.

Constrained node networks: Cyber security challenges

Constrained node networks face many challenges like mobility, reliability, scalability, management, availability,

interoperability and cyber security. And of all these challenges, cyber security is the most critical. Among all the aspects of it, securing data is very important as it determines both the confidentiality and integrity in the CIA (confidentiality, integrity and availability) triad. Conventional data security methods cannot be followed for constrained node networks because of resource constraints. The objective of this blog is to present the guidelines to securing data in constrained node networks in an IoT system.

Cyber security standards and the associated problems

There are various standards available to implement mechanisms for securing data. It is always good to follow these standards rather than to proceed with one's own custom solution, since it requires an enormous effort to understand the standards and select an appropriate one.

The list below covers a majority of the security standards applicable for embedded software development and its